Geospatial Analysis of Rome Taxi Trajectories

GPS Cleaning, Outlier Detection & Driver Analytics

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May 2025

Abstract

This report presents a geospatial data mining analysis of GPS trajectory data collected from Rome's taxi fleet during February and March 2014. The primary objectives were to clean and validate raw location records, compute fleet-wide and individual driver statistics, and characterise Driver 147's movement patterns relative to the overall fleet.

Raw data was processed through a stratified sampling pipeline (100 observations per driver) followed by spatial bound filtering and DBSCAN-based outlier detection. Of 31,500 sampled records, 28,889 were retained as valid. Fleet-wide geographic statistics revealed a mean operating location near Piazza Venezia, Rome's historical centre. Driver 147 exhibited a slightly below-average active duration of 602 hours and a wide geographic range spanning approximately 29.96 km diagonally across the city. These findings demonstrate the utility of density-based methods for GPS data quality assurance in urban mobility contexts.

1. Introduction

1.1 Background

Urban mobility data generated by GPS-equipped taxi fleets represents a rich source of spatiotemporal information. When properly mined, such data supports applications ranging from real-time dispatching and route optimisation to traffic management and urban planning. However, raw GPS data typically contains significant noise, including sensor errors, coordinate drift, and records logged outside the operational area, that must be addressed before meaningful analysis can proceed.

This report focuses on a dataset collected from Rome's taxi fleet across February and March 2014, covering hundreds of drivers and millions of location pings. The dataset is used to demonstrate a practical data cleaning and exploration pipeline using R.

1.2 Motivation

Fleet operators and city planners require accurate, clean mobility data to make evidence-based decisions. Uncleaned GPS data inflates geographic extents, distorts activity duration estimates, and introduces false patterns into any downstream analysis. Establishing a principled cleaning framework is therefore a prerequisite for any meaningful urban mobility study.

1.3 Objectives

- Perform stratified sampling to make the dataset computationally tractable for clustering.
- Remove geographically invalid GPS records using spatial bounding constraints.
- Identify and remove spatial outliers using DBSCAN density-based clustering.
- Compute fleet-wide geographic statistics (minimum, maximum, mean latitude and longitude).
- Determine the most active, least active, and average-active taxi drivers by operational time span.
- Conduct a deep-dive analysis of Driver 147: location statistics, activity duration, and route distance.

1.4 Scope

The analysis is bounded to GPS coordinates within Rome's defined bounding box: Latitude 41.80°–42.05°N, Longitude 12.35°–12.65°E. Driver 147 was identified via the Student–Taxi mapping file using student ID 8725111. Distance calculations use the Haversine formula applied to coordinate extremes as a bounding-box approximation.

2. Problem Statement

Raw GPS trajectory data from urban taxi fleets suffers from three categories of quality issues:

- Missing values, null or absent latitude/longitude entries due to signal loss.
- Out-of-bounds records, coordinates recorded outside the city boundary, caused by device errors or drivers temporarily leaving the operational area.
- Spatial outliers, geographically isolated points that are technically within the bounding box but are separated from the main cluster of activity, likely representing sensor noise or exceptional events.

Without addressing these issues, any derived statistics (means, ranges, activity durations) will be systematically biased. The problem is to design and implement a cleaning pipeline that removes all three categories of noise with minimal loss of genuine data, then use the cleaned dataset to characterise fleet behaviour and individual driver performance.

Key Question: How do Driver 147's geographic range and activity duration compare to the overall fleet, and what does this reveal about their operational patterns?

3. Data Description

3.1 Source & Format

The dataset (taxi.csv) contains GPS location records for a fleet of Rome taxis. Each row represents a single GPS ping for a specific driver at a specific timestamp.

Attribute	Detail
File	taxi.csv
Format	Comma-separated values (CSV) with header row
Key Fields	DriveNo (driver identifier), Date.and.Time, Latitude, Longitude
Raw Geographic Extent	Approx. 41.7°–42.35°N, 11.9°–13.1°E (before cleaning)
Valid Geographic Extent	Lat: 41.80°–42.05°N | Lon: 12.35°–12.65°E
Study Period	February – March 2014
Driver Mapping File	Student_Taxi_Mapping.txt, links Student ID 8725111 to Taxi/Driver 147

3.2 Sampling Strategy

The full dataset contains millions of records. To make DBSCAN computationally feasible while preserving representative coverage across drivers, stratified sampling was employed: 100 observations were drawn at random from each unique DriveNo. Drivers with fewer than 100 records were excluded. This produced a working sample of approximately 31,500 records.

Note: Driver 147's individual analysis uses the full dataset (all rows with DriveNo = 147) to maximise the fidelity of route and timing estimates, since the 100-record sample would be insufficient for meaningful trajectory visualisation.

3.3 Data Limitations

- Stratified sampling loses information about within-driver temporal patterns, gaps between trips, idle times, and shift structures cannot be recovered from 100 random samples.
- The Haversine distance is computed between coordinate extremes (bounding box corners), not along the actual route. This is a lower-bound estimate of total distance driven.
- No fare, passenger, or occupancy data is available, analysis is purely spatiotemporal.
- Timestamp precision and GPS device quality may vary across vehicles.

4. Methodology

4.1 Stratified Sampling

Using R's base sample() function, 100 records were drawn without replacement from each driver's full record set. Results were combined using do.call(rbind, sampled_list) into a single data frame. All sampled records were initially flagged as 'valid'.

4.2 Invalid Point Detection

A boolean index was constructed to flag records where latitude or longitude was NA, or where coordinates fell outside the Rome bounding box:

Latitude range: 41.80° N ≤ lat ≤ 42.05° N
Longitude range: 12.35° E ≤ lon ≤ 12.65° E
Flagged records receive status = 'invalid'

4.3 DBSCAN Outlier Detection

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) was applied to the longitude/latitude coordinate pairs of the stratified sample. Any point assigned to cluster 0 (noise) was reclassified as 'outlier'.

DBSCAN Parameter	Value (Stratified Sample) / Value (Driver 147 Full Set)
eps (neighbourhood radius)	0.01 degrees / 100 (unitless, larger to handle dense full dataset)
minPts (min neighbours)	2 / 10
Rationale	Small eps on sample captures tight geographic clusters; wider eps on full set avoids over-splitting dense routes

DBSCAN was chosen over IQR-based outlier detection because IQR was found to classify too many valid high-density points as outliers, and because DBSCAN naturally handles the spatial structure of GPS data where clusters are non-spherical and of varying density.

4.4 Activity Duration Computation

For each driver, the earliest and latest timestamps in the cleaned sample were identified using aggregate() with FUN = function(x) c(Max = max(x), Min = min(x)). Active span was computed as max(Date.and.Time) - min(Date.and.Time), converted to minutes and hours as required.

4.5 Haversine Distance (Driver 147)

The great-circle distance between Driver 147's minimum and maximum coordinate positions was calculated using the Haversine formula, providing an estimate of the geographic operational range:

R = 6,371,000 m

$$a = \sin^2(\Delta lat/2) + \cos(lat_1) \cdot \cos(lat_2) \cdot \sin^2(\Delta lon/2)$$
$$c = 2 \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{1-a})$$
$$d = R \cdot c$$

4.6 Tools & Libraries

Tool	Purpose
R (base)	Data loading, manipulation, statistical computation
dbscan (R package)	DBSCAN clustering for outlier detection
ggplot2 (R package)	Spatial scatter plots with colour-coded point types
base::aggregate	Per-driver min/max timestamp computation
Student_Taxi_Mapping.txt	Mapping of student ID 8725111 to Driver 147

5. Implementation

5.1 Pipeline Overview

The full data cleaning and analysis pipeline follows eight sequential steps:

- Step 1, Load: read.csv('taxi.csv', header=TRUE, sep=',', dec='.')
- Step 2, Stratified Sample: Loop over unique(taxi_data$DriveNo), sample 100 rows, bind with do.call(rbind, ...).
- Step 3, Flag Invalids: Boolean index on coordinate bounds → status = 'invalid'.
- Step 4, DBSCAN: dbscan(coordinates, eps=0.01, minPts=2) → cluster==0 → status = 'outlier'.
- Step 5, Clean: subset(stratified_sample, status != 'outlier' & status != 'invalid').
- Step 6, Fleet Stats: min/max/mean on cleaned_data$Latitude and $Longitude.
- Step 7, Activity: aggregate() by DriveNo → time difference → most/least/average active.
- Step 8, Driver 147: Reload full dataset → filter DriveNo==147 → re-clean → compute stats and Haversine distance.

5.2 Visualisations Produced

Two ggplot2 scatter plots were generated for the stratified sample:

- Pre-cleaning plot: All points colour-coded as valid (green), invalid (purple), or outlier (red), with Rome's bounding box and city centre marked.
- Post-cleaning plot: Valid points only (blue), with Rome centre marker, confirms spatial concentration around the city centre.

A third plot shows Driver 147's full cleaned trajectory across Rome, revealing the driver's route network as a dense web of blue points covering most of the bounding box area.

6. Results

6.1 Data Cleaning Summary

28,889	2,536	75
Valid Points Retained	Invalid (Out of Bounds)	DBSCAN Outliers Removed

Of the 31,500 stratified sample records, 8.1% were removed as spatially invalid and 0.24% were removed as DBSCAN outliers. The retained 28,889 points form a coherent geographic cluster concentrated within Rome's urban core.

6.2 Fleet-Wide Geographic Statistics

Statistic	Value
Minimum Latitude	41.80004° N
Maximum Latitude	42.03442° N
Mean Latitude	41.89835° N → approx. Piazza Venezia / historic centre
Minimum Longitude	12.35006° E
Maximum Longitude	12.64675° E
Mean Longitude	12.48462° E
Lat range covered	0.234° (approx. 26 km north-south)
Lon range covered	0.297° (approx. 23 km east-west)

The mean location (41.898°N, 12.485°E) falls near Piazza Venezia, historically the geographic and commercial heart of Rome. This is consistent with expectations for a taxi fleet primarily serving commuters, tourists, and business travellers in the city centre.

6.3 Driver Activity Analysis

Activity Metric	Value
Most Active Driver	43,145 minutes (≈ 719 hours across study period)
Least Active Driver	13.8 minutes (likely very short or incomplete record)
Average Active Span	37,256 minutes (≈ 621 hours)
Driver 147 Span	36,134 minutes (≈ 602 hours) , slightly below average

The extreme range between the most and least active drivers (719 hours vs 0.23 hours) suggests significant heterogeneity in the fleet, likely a mix of full-time drivers with continuous operation, part-time drivers, and vehicles whose GPS devices were logged only briefly. Driver 147 at 602 hours falls 3% below the fleet mean, indicating slightly below-average participation over the study period.

6.4 Driver 147, Individual Profile

602 hrs	**29.96 km**	**41.914°N**
Active Duration	Geographic Range	Mean Latitude

Driver 147 Metric	Value
Min Latitude	41.8026° N
Max Latitude	42.0500° N (at northern bounding limit)
Mean Latitude	41.9142° N (0.016° north of fleet mean)
Min Longitude	12.3501° E (at western bounding limit)
Max Longitude	12.6493° E (at eastern bounding limit)
Mean Longitude	12.4572° E (0.027° west of fleet mean)
Active Duration	≈ 602.2 hours (36,134 minutes)
Geographic Range (Haversine)	≈ 29,960 m (29.96 km bounding diagonal)

Driver 147's coordinates span nearly the full bounding box in both dimensions, indicating wide-area coverage rather than specialised zone operation. The mean latitude is 0.016° north of the fleet average, suggesting a modest preference for northern Rome routes (e.g., Parioli, Flaminio, or Vatican areas). The Haversine bounding diagonal of 29.96 km represents the maximum geographic extent of the driver's operating area.

7. Discussion

7.1 Effectiveness of the Cleaning Pipeline

The combined approach of spatial bounding and DBSCAN proved effective. IQR-based outlier detection was considered but rejected because it classified too many legitimate high-density urban records as outliers. DBSCAN, with eps=0.01 degrees, successfully isolated genuinely isolated points without over-removing dense urban clusters.

The 8.1% invalid rate (2,536 points) is notable, it suggests that a meaningful proportion of GPS pings are logged outside the city, possibly during vehicle maintenance trips, repositioning runs, or GPS signal acquisition delays when devices first activate.

7.2 Fleet vs Driver 147 Comparison

Driver 147 sits close to the fleet average on most metrics, making them a representative driver rather than an outlier. The slightly northern mean latitude and near-complete bounding box coverage suggest this driver may specialise in longer cross-city trips rather than concentrated short-distance fares in a single district.

The Haversine distance of 29.96 km is a bounding-box diagonal, not a route distance. Actual kilometres driven would be substantially higher. A proper route distance calculation would require sequentially computing inter-point Haversine distances along the chronologically ordered trajectory, a logical next step.

7.3 Sampling Limitations

Using 100 random samples per driver introduces variance, particularly for drivers with highly non-uniform activity patterns (e.g., those who concentrate trips in specific time windows). The asymmetry between the stratified sample (used for fleet stats) and the full Driver 147 dataset (used for individual stats) means direct numerical comparisons should be interpreted cautiously, the difference in activity duration between Driver 147 and the fleet mean is partly an artefact of this asymmetry.

8. Conclusion

This analysis successfully implemented a robust GPS data cleaning pipeline for Rome's taxi fleet dataset. Stratified sampling, spatial bound filtering, and DBSCAN outlier detection together retained 28,889 valid records from an initial 31,500-record sample, a retention rate of 91.7%.

Fleet-wide statistics confirm that taxi activity is concentrated around Rome's historical centre, with mean coordinates near Piazza Venezia. Driver 147 (mapped to student ID 8725111) demonstrated slightly below-average activity duration (602 vs 621 hours fleet mean) and wide geographic coverage spanning virtually the entire city bounding box, with a diagonal operational range of approximately 29.96 km.

The objectives of Task 1 were fully achieved. The cleaned dataset provides a reliable foundation for further analysis including route modelling, demand forecasting, and driver productivity benchmarking.

9. Recommendations

- Increase stratified sample size to 500 observations per driver to improve statistical stability, particularly for activity duration estimates.
- Compute sequential Haversine distances along the chronologically ordered trajectory for accurate total kilometres driven rather than bounding-box diagonal.
- Segment trips into 'occupied' and 'empty' legs using speed and pause patterns to compute productive vs idle time per driver.
- Apply kernel density estimation (KDE) to the cleaned fleet data to visualise hotspots and guide dispatching strategy.
- Investigate whether drivers with very low activity spans (< 100 minutes) represent genuine part-time drivers or data artefacts, and consider excluding them from fleet benchmarks.
- Incorporate time-of-day and day-of-week analysis to identify peak activity windows across the fleet.

10. Limitations & Future Work

10.1 Data Limitations

- Stratified sampling destroys temporal continuity, trip boundaries, shift patterns, and idle times cannot be recovered from 100 random samples per driver.
- No occupancy, fare, or passenger data, purely spatiotemporal analysis.
- GPS device quality and sampling frequency vary across vehicles, introducing non-uniform spatial resolution.

10.2 Method Limitations

- DBSCAN parameters (eps=0.01, minPts=2) were chosen heuristically. A systematic grid search with silhouette analysis would improve parameter selection.
- Haversine bounding diagonal substantially underestimates true route distance.

10.3 Future Work

- Apply LSTM recurrent networks to model sequential GPS trajectories for route prediction.
- Perform trip segmentation using speed thresholds and time gaps to distinguish individual fare trips.
- Develop an interactive Shiny dashboard for fleet managers to explore driver statistics in real time.
- Cross-reference taxi GPS data with Rome's traffic and event data to understand demand drivers.

Report: Task 1, Geospatial Analysis of Rome Taxi Trajectories
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Institution: University of Wollongong | Date: May 2025
Tools: R, dbscan, ggplot2